DATA COLLAGE · V1.0

RECORD · RENDER · REANIMATE

18 Skills.
One Substrate.
Recorded, rendered, reanimated.

A statistical collage of the SEARM + Photoreal Cartographic Native skill packs — the cryptographic/epistemic discipline and the rendering side, together forming the VVU digital twin substrate.

RECORD → RENDER → REANIMATE → RECORD AGAIN

18 skills · 4,947 lines · 35,768 words
236 trigger keywords · 73 cross-references · 173 checklist items
SEARM pack: 12 skills · Photoreal pack: 6 skills

18	4,947	236	73
SKILLS	LINES	TRIGGERS	X-REFS

01 · STATISTICAL DASHBOARD

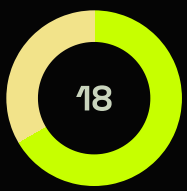
Pack Statistics

Aggregate metrics across the 18-skill substrate

TOTAL SKILLS	TOTAL LINES	TOTAL WORDS
18	4,947	35,768
12 SEARM + 6 Photoreal	avg 274 per skill	avg 1,987 per skill

TRIGGER KEYWORDS	CROSS-REFERENCES	CHECKLIST ITEMS
236	73	173
avg 13 per skill	inter-skill links	verification gates

PACK DISTRIBUTION



SEARM Pack	12 skills · 2,836 lines
Photoreal Cartographic Native Pack	6 skills · 2,111 lines

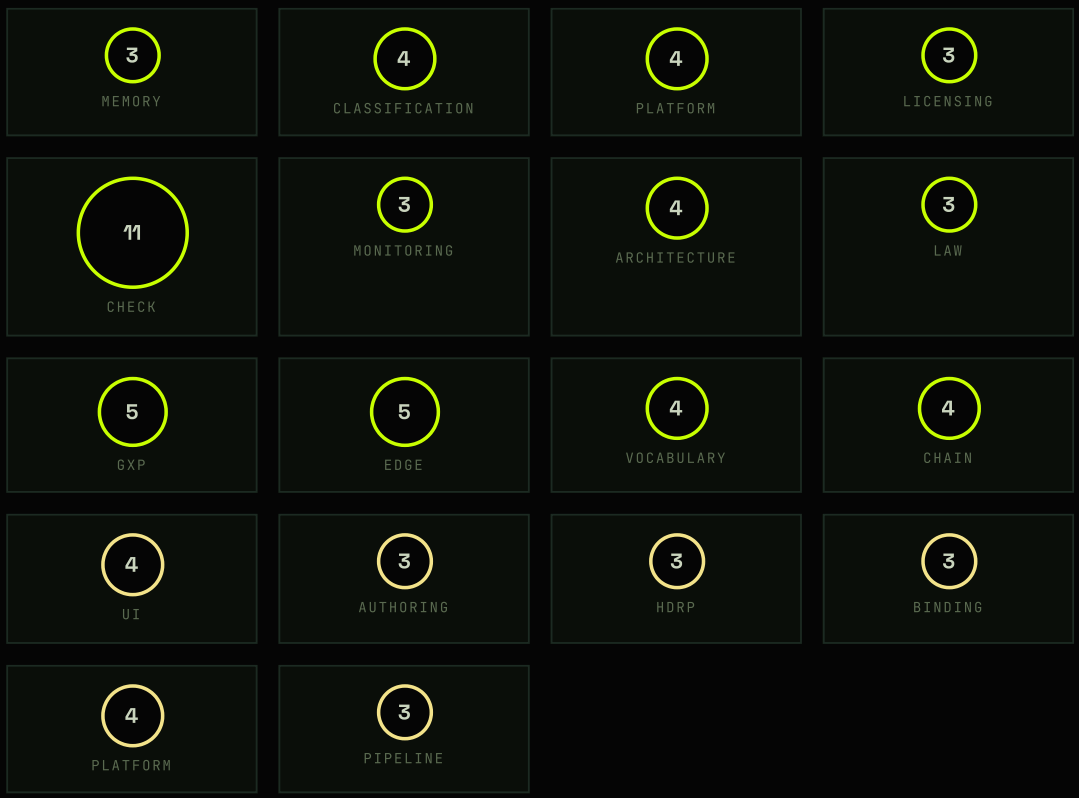
TOP 15 TRIGGER KEYWORDS (BY FREQUENCY ACROSS SKILLS)



02 · CROSS-REFERENCE NETWORK

Skill Interconnects

Each node sized by cross-reference count · lime = SEARM · ochre = Photoreal



NETWORK INSIGHT

The SEARM Estate Drift Check (Skill 12) is the master gate — it references all 11 other SEARM skills. The WORM Chain (Skill 3) and No-Primitives Law (Skill 1) are the most-referenced foundational skills. The Photoreal pack cross-references internally (HDRP ↔ Sketchfab pipeline ↔ Material authoring) and ties to the SEARM pack through the local-first and cross-platform skills.

03 · SKILL COLLAGE

The 18 Skills

Each card shows the skill's stats, trigger keywords, and one-liner

● Below: all 18 skills rendered as data cards. Each card captures the skill's name, pack, line/word/trigger counts (with normalized bars), the top trigger keywords, and the skill's one-liner. This is the **RECORD** phase — each skill's state captured as a statistical snapshot.

01 searm-agent-memory

SEARM PACK

236	11	3
LINES	TRIGGER	XREF

Implements the SEARM AI agent memory system — file-based Markdown with frontmatter, progressive loading, two distinct scopes (read-only org conventions vs read-write team/agent mem...



TRIGGERS: agent memory, context file, MEMORY.md, frontmatter, content hash, precondition *+5 more*

→ Memory is files. Files are diffable, portable, attributable, reversible. The agent loads what it needs, when it needs it — never the whole store.

02 searm-claim-classification

SEARM PACK

250	12	4
LINES	TRIGGER	XREF

Enforces the SEARM evidence assessment and claim classification discipline. Every number or claim produced by the system must be tagged [MEASURED], [PUBLIC: source], [DEMO], or [VE...



TRIGGERS: deployed for, pilot in discussion, deployed for, pilot, in production, live at *+6 more*

→ Every number carries its epistemic status. "Deployed for" is a measured claim — and unmeasured, it is forbidden.

03 searm-cross-platform

SEARM PACK

227	13	4
LINES	TRIGGER	XREF

Enforces the SEARM mobile-first + desktop-native cross-platform engineering constraint. SEARM must run on mobile (Android via Termux, iOS via Shortcuts/HealthKit, PWA) and desktop-...



TRIGGERS: Tauri, Electron, Termux, PWA, HealthKit, TrueDepth *+7 more*

→ Take the install artefact. Put it on a clean machine with no internet. Install it. Open a `chain.json`. Run the verifier. Does it produce `✅ CHAIN INTACT`?

04 searm-dual-licensing

SEARM PACK

238	13	3
LINES	TRIGGER	XREF

Enforces the SEARM open-source + commercial dual-licensing discipline. The SEARM core (verifier, chain, EIS library, IVE dashboard) is open-source and auditable; the enterprise wra...



TRIGGERS: GPL, Apache, MIT, AGPL, CC-BY, open-source *+7 more*

→ The open-source core is sufficient to produce and verify receipts. The commercial wrappers add convenience, integrations, and brand, never the cryptographic substance.

05 searm-estate-drift-check

SEARM PACK

220 LINES | 10 TRIGGER | 11 XREF

The final review gate for SEARM — every new feature, artifact, code change, or claim must pass the Estate/Narrative Drift Check before it ships. The check asks one question: does t...



TRIGGERS: `it changes what is physically true, drift check, estate, narrative, review, before merge` *+4 more*

→ Does this change what is physically true, or only how what is physically true is shown?

06 searm-fail-closed-monitoring

SEARM PACK

304 LINES | 14 TRIGGER | 3 XREF

Implements the SEARM fail-closed timeout and sensor monitoring loop — the 15-second fail-closed timeout, the 60-second heartbeat, the FAIL_CLOSED / SENSOR_ABSENT / LEAK_ACTIVE / BR...



TRIGGERS: `fail-closed, fail closed, 15 seconds, 60 seconds, heartbeat, SENSOR_ABSENT` *+8 more*

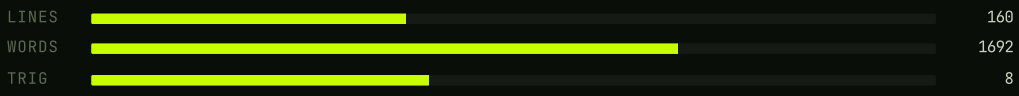
→ Absence is a receipt, not an error. The chain records the silence as carefully as it records the noise.

07 searm-local-first-architecture

SEARM PACK

160 LINES | 8 TRIGGER | 4 XREF

Enforces the SEARM Not-SaaS, local-first architecture constraint. The system's core engine, WORM ledger, and verifier must operate entirely offline and locally on mobile and deskto...



TRIGGERS: `local-first, Not-SaaS, offline verification, no phone-home, no API keys, verifier` *+2 more*

→ Take the install artefact (the `.apk`, the `.dmg`, the `.exe`, the standalone binary, the static HTML+JS bundle). Put it on a machine with no network. Install it. Open a `chain.json` file. Run the verifier. Does it produce `CHAIN INTACT: n entries verified`?

08 searm-no-primitives-law

SEARM PACK

147 LINES | 7 TRIGGER | 3 XREF

Enforces the SEARM No-Primitives Law — the foundational constraint that the agent must never guess, interpolate, or fabricate physical data. Use this skill whenever the agent is re...



TRIGGERS: `the estate, No-Primitives, fail closed, SENSOR_ABSENT, the estate, the narrative` *+1 more*

→ Does this code change what is physically true, or does it only change how what is physically true is shown?

09 searm-popia-gxp

SEARM PACK

306 LINES | 16 TRIGGER | 5 XREF

Enforces SEARM POPIA, consent, and GxP boundary compliance. The ledger holds salted hashes only — never raw PHI or names. Consent receipts are explicitly recorded. For regulated en...



TRIGGERS: `POPIA, GDPR, HIPAA, GxP, GMP, FDA` *+10 more*

→ The ledger proves what was measured, never who was measured. Consent is a receipt. In regulated environments, SEARM sits beside the validated system, never inside it.

10 searm-read-only-edge

SEARM PACK

259 LINES | 14 TRIGGER | 5 XREF

Enforces the SEARM read-only edge integration discipline for OT/IT safety. The edge node component must never write to, control, or interlock with PLCs, robots, BMS, MES, or any cl...

LINES 259
WORDS 2281
TRIG 14

TRIGGERS: edge node, PLC, Modbus, OPC-UA, DNP3, BMS *+8 more*

→ SEARM observes. SEARM proves. SEARM does not control. The SCADA does not know we exist; we know the SCADA only through what we read.

11 searm-vertical-vocabulary

SEARM PACK

234 LINES | 13 TRIGGER | 4 XREF

Enforces the SEARM vertical skinning discipline and exact string vocabulary. One engine powers multiple verticals (Health, Finance, Industrial, Infrastructure, Community/Gov); each...

LINES 234
WORDS 1807
TRIG 13

TRIGGERS: vertical, skin, ENGINE, DRIVETRAIN, LIVE, SIM *+7 more*

→ One engine, many skins. The skin changes the narrative; the estate — the chain, the verifier, the math — stays constant.

12 searm-worm-chain

SEARM PACK

255 LINES | 10 TRIGGER | 4 XREF

Implements the SEARM WORM (Write Once, Read Many) cryptographic receipt chain — the canonical hash chain that proves physical state changes. Implements RFC 8785 JCS canonicalisatio...

LINES 255
WORDS 1555
TRIG 10

TRIGGERS: WORM, chain.json, verify-chain, Ed25519, RFC 8785, RFC 3161 *+4 more*

→ An auditor takes the `chain.json` file, runs `node verify-chain.js`, and gets `✅ CHAIN INTACT: 4,217 entries verified`. No phone call to SEARM. No API key. No trust. Just math.

13 cartographic-realistic-ui

PHOTOREAL PACK

431 LINES | 15 TRIGGER | 4 XREF

Building cartographic, photorealistic UIs with Mapbox GL JS / MapLibre GL JS — covering 3D terrain (Mapbox Terrain-DEM v1), satellite imagery, vector tiles, custom 3D building extr...

LINES 431
WORDS 2200
TRIG 15

TRIGGERS: command-center, Mapbox, MapLibre, vector tiles, raster DEM, 3D terrain *+9 more*

→ A twin does not exist in a void. It exists at a place. The cartographic UI grounds the twin in its real terrain, with its real satellite imagery, at its real coordinates — so the viewer believes "this is here".

14 photoreal-material-authoring

PHOTOREAL PACK

349 LINES | 16 TRIGGER | 3 XREF

Authors photorealistic PBR materials — surface textures that make a 3D model look like real metal, plastic, skin, concrete. Covers the metallic-roughness workflow, capture-based te...

LINES 349
WORDS 2440
TRIG 16

TRIGGERS: PBR, metallic-roughness, albedo, roughness map, normal map, AO *+10 more*

→ A photoreal twin is built from photoreal materials. The material is what makes the viewer believe "this is real metal, real paint, real skin" — and belief is the entire epistemic point.

15 photoreal-unity-hdrp

PHOTOREAL PACK

253 LINES | 10 TRIGGER | 3 XREF

Authoring Unity HDRP projects for photorealistic digital twins and physical-environment-aware native applications. Covers HDRP setup, ray-traced lighting, volumetric fog, post-proc...

LINES		253
WORDS		2173
TRIG		10

TRIGGERS: Unity HDRP, ray tracing, path tracing, volumetrics, Probe Volume, Decal Projector *+4 more*

→ Photorealism is an epistemic signal: if the twin looks like a render, the viewer does not trust the data. If it looks like a photograph, they do.

16 physical-environment-binding

PHOTOREAL PACK

342 LINES | 16 TRIGGER | 3 XREF

Binds the digital twin to the real physical environment — GPS, compass, ambient light, depth camera, IMU, weather, time-of-day sun, ARKit/ARCore spatial tracking, live sensor strea...

LINES		342
WORDS		2203
TRIG		16

TRIGGERS: ARKit, ARCore, ambient light, compass, GPS, IMU *+10 more*

→ A twin that ignores its physical context is a render. A twin that responds to its physical context is a presence — and presence is what makes the viewer believe.

17 seamless-native-cross-platform

PHOTOREAL PACK

412 LINES | 18 TRIGGER | 4 XREF

Delivers photorealistic digital twins as seamless native applications across desktop (Windows/macOS/Linux via Tauri) and mobile (iOS/Android via Capacitor or React Native WebView) ...

LINES		412
WORDS		2270
TRIG		18

TRIGGERS: Tauri, Electron, Capacitor, React Native, Expo, native build *+12 more*

→ Take the install artefact. Put it on a machine with no internet. Install it. Open a `chain.json` file. Run the verifier. Does it produce `✅ CHAIN INTACT`? Open the twin. Does it render with cached assets?

18 sketchfab-3d-pipeline

PHOTOREAL PACK

324 LINES | 20 TRIGGER | 3 XREF

The 3D modelling pipeline that produces photorealistic, Sketchfab-publishable, SEARM-grade digital twin assets — covering topology discipline, UV unwrapping, PBR material authoring...

LINES		324
WORDS		2334
TRIG		20

TRIGGERS: GLB, GLTF, FBX, OBJ, Sketchfab, PBR materials *+14 more*

→ A photoreal twin is built from photoreal assets. The pipeline — blockout, sculpt, retopo, UV, bake, PBR, LOD, GLB, Sketchfab — turns a scan or a concept into a runtime-ready, attribution-tracked, license-clean asset.

04 · THE LOOP

Record • Render • Reanimate

The cyclical discipline that keeps the skills substrate alive



THE LOOP · EXPLAINED

01 RECORD — Each skill's state is captured as a Markdown file with frontmatter (per SEARM Skill 6 — Agent Memory). The content hash is recorded; the write is preconditioned.

02 RENDER — The skills are visualized as a data collage (this PDF). Statistics, trigger keywords, cross-references, and one-liners are rendered as a statistical surface.

03 REANIMATE — When new SEARM work happens (a new twin, a new vertical, a new pilot), the skills re-trigger. The agent consults the relevant skills; the skills shape the work.

04 RECORD AGAIN — The new work is appended to the WORM chain (per SEARM Skill 3). The chain is the persistent record; the skills are the active discipline. The loop closes; the cycle begins again.

THE ONE-LINER

The skills are the discipline. The chain is the record. The collage is the render. The loop is the life.